

# NAS Powered Deep Image Prior for Electrical Impedance Tomography

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**Abstract**—In this paper, we introduce a novel approach that combines neural architecture search (NAS) with the deep image prior (DIP) framework for electrical impedance tomography (EIT) reconstruction. Deep neural networks have proven effective as DIPs in various image reconstruction tasks, but the appropriate prior is task-dependent. Manually designing network architectures for EIT reconstruction is challenging. Our method automates this process by using NAS to identify optimal neural network configurations tailored for EIT reconstruction. This approach eliminates the need for rare labeled data, which is a significant advantage in EIT applications. Extensive validation using both simulated and experimental data showcases the effectiveness of our NAS-powered DIP approach. Comparative evaluations against traditional methods and state-of-the-art techniques consistently demonstrate superior reconstruction results and robustness against noise. Our approach opens up exciting possibilities for advancing EIT reconstruction methods, with potential applications in medical imaging and industrial testing.

**Index Terms**—Neural architecture search (NAS), deep image prior (DIP), unsupervised learning, image reconstruction, electrical impedance tomography.

## I. INTRODUCTION

CONVOLUTIONAL Neural Networks (CNNs) have emerged as a cornerstone in the realm of image processing and image reconstruction. These specialized neural networks are meticulously crafted to excel in the analysis of visual content or measurement data, proving instrumental in a multitude of applications, ranging from image denoising [1], [2], [3] and

super-resolution [4], [5], [6] to the domain of medical image reconstruction [7], [8], [9]. They have essentially become a linchpin for discerning valuable information from visual content or measurement data.

In sharp contrast to the resource-intensive prerequisites of supervised learning methods, unsupervised learning has emerged as a widely embraced paradigm within the expansive ambit of deep learning. It provides an alternative avenue for generating meaningful inferences in the complex domain of inverse problems without relying on labeled data. A notable exemplar of unsupervised learning is the Cycle Consistent Generative Adversarial Network (CycleGAN) [10], a versatile framework with applications spanning diverse domains. However, traditional unsupervised methods confront inherent challenges, including the reliance on manually crafted priors, presumptions concerning image statistics, and the intricate task of capturing sophisticated image structures [11].

Within the landscape of image processing and reconstruction, the Deep Image Prior (DIP) [12] framework has emerged as a powerful unsupervised approach. DIP has unveiled a transformative insight: that network architecture itself serves as a pivotal prior for image reconstruction. This groundbreaking approach harnesses the latent knowledge embedded within deep neural networks to undertake image processing tasks. As exemplified by the DIP framework, Ulyanov et al. [12] embarked on a journey by employing random noise as input and fitting a randomly initialized network to corrupted images, ultimately yielding high-quality images. What sets DIP apart is its unique characteristic of circumventing the traditional training process, making DIP-based methods widely applicable in image reconstruction, especially in the intricate domain of medical image reconstruction.

Moreover, considerable efforts have been dedicated to amplifying the performance of DIP-based methods. For instance, the introduction of DeepRed [13] served as a means to enhance DIP by integrating existing denoising techniques for the regularization of inverse problems. Baguer et al. [14] innovatively merged the DIP approach with classical regularization techniques for computed tomography (CT) reconstruction. Shan et al. [15] harnessed the power of DIP to generate positron emission tomography (PET) reconstructions from incomplete data. The DIP-based DeepEIT framework [16] was recently developed to facilitate electrical impedance tomography (EIT) reconstruction, yielding promising results without the need for labeled data. The key idea of DeepEIT is to utilize an encoder-decoder network to parameterize the conductivity distribution.

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Although DIPs have found wide-ranging applications in numerous domains, but their true potential hinges on the intricacies of network architecture design. A well-crafted architecture in DIPs can yield exceptional performance; however, the manual development of such architectures remains a formidable challenge for researchers, fraught with complexities. On one hand, this process heavily relies on the prior knowledge and expertise of researchers in the respective field. On the other hand, designing an architecture is a time-consuming and error-prone endeavor. Canonical network architectures like AlexNet [17], VggNet [18], ResNet [19], and U-net [20] have been meticulously designed by seasoned experts, underscoring the difficulty of crafting novel architectures.

Consequently, an increasing number of researchers are turning their attention to NAS as a potential solution [21]. NAS endeavors to automatically create neural architectures that harness limited computational resources to achieve optimal performance with minimal human intervention [22], [23]. The inception of NAS is commonly attributed to the work of Zoph and Le [24], who employed reinforcement learning algorithms to search for effective neural architectures. Subsequently, Baker et al. combined Q-learning and meta-modeling techniques to tackle image classification tasks, surpassing prior meta-modeling approaches [25]. Furthermore, large-scale evolution models, utilizing evolutionary algorithms to minimize human involvement, have demonstrated promising results in image classification [26]. These studies underscore the viability of the NAS approach.

In light of the potential for NAS to enhance DIP-based methods by discovering well-suited network architectures, Chen et al. [27] introduced NASDIP to address image processing tasks. NASDIP subdivided the search space into the design of upsampling cells and cross-level feature connections within the standard U-net architecture. Through exploration within this search space and the application of reinforcement learning techniques, NASDIP unearthed a network architecture surpassing the performance of conventional designs for specific tasks such as image denoising and image super-resolution, among others.

Building upon the achievements of the aforementioned works, we employ the NASDIP framework to address EIT reconstruction challenges and enhance the DeepEIT framework [16] using a model selection technique without the need for extensive training. To the best of our knowledge, this work is the first to apply NASDIP for image reconstruction and to do so without supervision, leveraging a state-of-the-art perceptual metric-power spectral density (PSD) distance to guide the search for optimal architectures. To assess the performance of our method, we conduct validations and robustness studies using simulation data and experimental water tank data. The results underscore the exceptional performance and robustness of our method, even in the presence of low signal-to-noise ratio (SNR) noise. The primary contributions of this work can be summarized as follows:

- *Innovative Integration*: Pioneering the integration of NAS with DIP for enhanced Electrical Impedance Tomography (EIT) reconstruction.
- *Autonomous Architecture Selection*: Introducing a method proficient in autonomously searching for optimal neural

architectures, tailoring the performance of DIP to address varied challenges in EIT reconstruction.

- *Less Reliance on Early-Stopping*: Reducing dependence on early-stopping and demonstrating robustness in the selection of iteration numbers, overcoming a bottleneck for original DIP that typically varies with tasks and architectures.
- *Superior Performance*: Extensive evaluation through simulation and experimental tests, revealing superior performance compared to traditional methods and original DIP-based approaches in EIT reconstruction tasks.
- *Robustness Against Noise*: Showcasing the method's robustness against noise, particularly in experimental settings, highlights its capacity to maintain robust performance in real-world scenarios.

The structure of this paper is as follows. We briefly review EIT forward problems in Section II. Section III introduces the proposed EIT reconstruction framework, followed by implementation details in Section IV. Reconstruction results using simulation and experiment data are presented in Section V. Section VI analyzes ablation and robustness studies. Lastly, Section VII offers concluding remarks and summarizes key findings.

## II. FORWARD MODEL OF EIT PROBLEM

The EIT forward model is to calculate the boundary measurement  $V$  from a given conductivity distribution  $\sigma$  and the injected currents  $I$  in the imaging domain  $\Omega$ . To inject the currents and measure the corresponding voltages,  $L$  electrodes denoted as  $e_1, e_2, \dots, e_L$  are equidistantly placed around the domain boundary  $\partial\Omega$ . The conductivity and electric potential on position  $m = (x, y) \in \Omega$  are respectively denoted as  $\sigma(m)$  and  $u(m)$ . For modeling these measurements, we use the so-called complete electrode model (CEM), which consists the following equations:

$$\nabla \cdot (\sigma(m) \nabla u(m)) = 0, \quad m \in \Omega, \quad (1)$$

$$u(m) + z_i \sigma(m) \frac{\partial u(m)}{\partial n} = U_i, \quad m \in e_i, \quad i = 1, \dots, L, \quad (2)$$

$$\int_{e_i} \sigma(m) \frac{\partial u(m)}{\partial n} dl = I_i, \quad i = 1, \dots, L, \quad (3)$$

$$\sigma(m) \frac{\partial u(m)}{\partial n} = 0, \quad m \in \partial\Omega \setminus \bigcup_{i=1}^L e_i. \quad (4)$$

Here,  $n$  is the outward unit normal of the boundary,  $I_i$  and  $U_i$  are the injected current and electric potential on the  $i$ th electrode, and  $z_i$  is the contact impedance between the  $i$ th electrode and boundary.

Furthermore, to ensure a unique solution, the currents should satisfy the charge conservation law and a ground level for the potentials should be fixed:

$$\sum_{l=1}^L I_l = 0, \quad (5)$$

$$\sum_{l=1}^L U_l = 0. \quad (6)$$

Finally, using the finite element method (FEM) to numerically approximate the CEM in (1)–(6) and modeling the measurement errors as additive Gaussian noise, the EIT observation model is represented as follows:

$$V = U(\sigma) + \eta. \quad (7)$$

Here,  $V$  is a vector consisting of the measurements,  $U(\sigma)$  is the forward solution based on the given conductivity distribution  $\sigma$ , and  $\eta$  is an additive Gaussian noise.

### III. NAS POWERED DIP FOR EIT

In this section, we start by briefly introducing the original DIP-based EIT framework. Then we discuss the necessity of neural architecture search in DIP and the corresponding evaluation method during architecture selection process. Finally, we illustrate the NASDIP-based reconstruction framework for EIT.

#### A. DeepEIT: Deep Image Prior Enabled EIT

Expanding on the achievements of DIP, Liu and colleagues introduced a reconstruction framework entitled *DeepEIT* in the recent work [16]. DeepEIT employs an encoder-decoder network as an implicit prior for EIT reconstruction. Within DeepEIT, the encoder-decoder network takes a random noise input denoted as  $z$  and produces the reconstructed conductivity distribution  $\sigma$  as its output. Mathematically, the network is treated as a parameterization function represented by  $f_\theta$ , where  $\theta$  represents the network's parameters. Since the EIT measurement domain is usually non-rectangular, it becomes necessary to establish a mapping projection denoted as  $P$  that connects the measurement domain to the rectangular image domain. This enables the characterization of the conductivity distribution as follows:

$$\sigma = P(f_\theta(z)). \quad (8)$$

Subsequently, by utilizing the observation model in (7), the loss function  $L(\theta)$  can be expressed as

$$L(\theta) = \|V - U(P(f_\theta(z)))\|^2. \quad (9)$$

Considering the severely ill-posed nature of EIT, regularization techniques, such as total variation, are often applied to enhance its conditioning. Consequently, the updated loss function is expressed as:

$$L(\theta) = \|V - U(P(f_\theta(z)))\|^2 + \alpha TV(f_\theta(z)). \quad (10)$$

Here, the parameter  $\alpha$  signifies the weight assigned to the total variation (TV) regularization term, which is defined as

$$TV(\hat{\sigma}) = \sum_i S_i \sqrt{(d_x \hat{\sigma})_i^2 + (d_y \hat{\sigma})_i^2} + \beta, \quad \hat{\sigma} = f_\theta(z). \quad (11)$$

Here,  $i$  represents the  $i$ th element,  $S_i$  denotes the area of the  $i$ th element, and  $d_x$  and  $d_y$  are the differential operators along the

x-axis and y-axis, respectively. To ensure the differentiability of the TV regularization term, a small positive value  $\beta$  is assigned.

Despite DeepEIT demonstrating impressive results, it's crucial to emphasize that the network architecture in DeepEIT is custom-designed. This process involves the creation of different architectures to derive optimal priors specific to particular problems. However, as deep learning continues to evolve, crafting efficient neural networks becomes increasingly intricate. As a response to this challenge, we apply the concept of NAS in the following subsection to enhance DeepEIT.

#### B. Neural Architecture Search

NAS entails an automated approach to designing the structure of a neural network. Taking inspiration from the recent work in NASDIP [27], we adopt a similar search space to boost the DeepEIT framework. Within the context of encoder-decoder networks, the key challenge lies in efficiently recovering feature maps using the features extracted by encoders. Therefore, our search space predominantly encompasses two vital components: upsampling cells and skip connections.

The upsampling cells are further subdivided into five components, encompassing methods for altering spatial resolution, feature transformation techniques, kernel size, dilation rate, and activation functions. Combining these operators allows us to readily explore diverse designs for upsampling cells. Additionally, we search for skip connections that connect encoders and decoders to leverage features extracted in the encoders (e.g., from the 1st feature level in the encode to 5th feature level in the decoder). To illuminate the fundamental concept, we have visually depicted the main flow of NAS in the upper segment of Fig. 1. A more detailed implementation can be found in Section IV-A.

#### C. Neural Architecture Evaluation

Whereas NAS technology can produce a diverse array of networks, it's crucial to acknowledge that evaluating models for their performance through iteration is a time-consuming task. Arican et al. demonstrated in [28] that the power spectral density (PSD) distance between the network output without iteration and a corrupted image can act as an indicator of the model's capability to achieve enhanced performance. To minimize time expenditure, we also employ PSD distance as a metric to choose the optimal architecture.

Mathematically, PSD serves as a tool employed to characterize the spectral properties of a signal [29]. When it comes to fine textures and coarse textures, the PSD tends to primarily highlight high and low frequencies, respectively. To illustrate the detailed computation of the PSD, we initially present the formula for the two-dimensional discrete Fourier transform  $F(v_x, v_y)$  of the space domain function  $f(x, y)$ :

$$F(v_x, v_y) = \sum_{n_x=0}^{N_x-1} \sum_{n_y=0}^{N_y-1} f(x, y) e^{i2\pi(v_x x/N_x + v_y y/N_y)} \Delta v_x \Delta v_y. \quad (12)$$

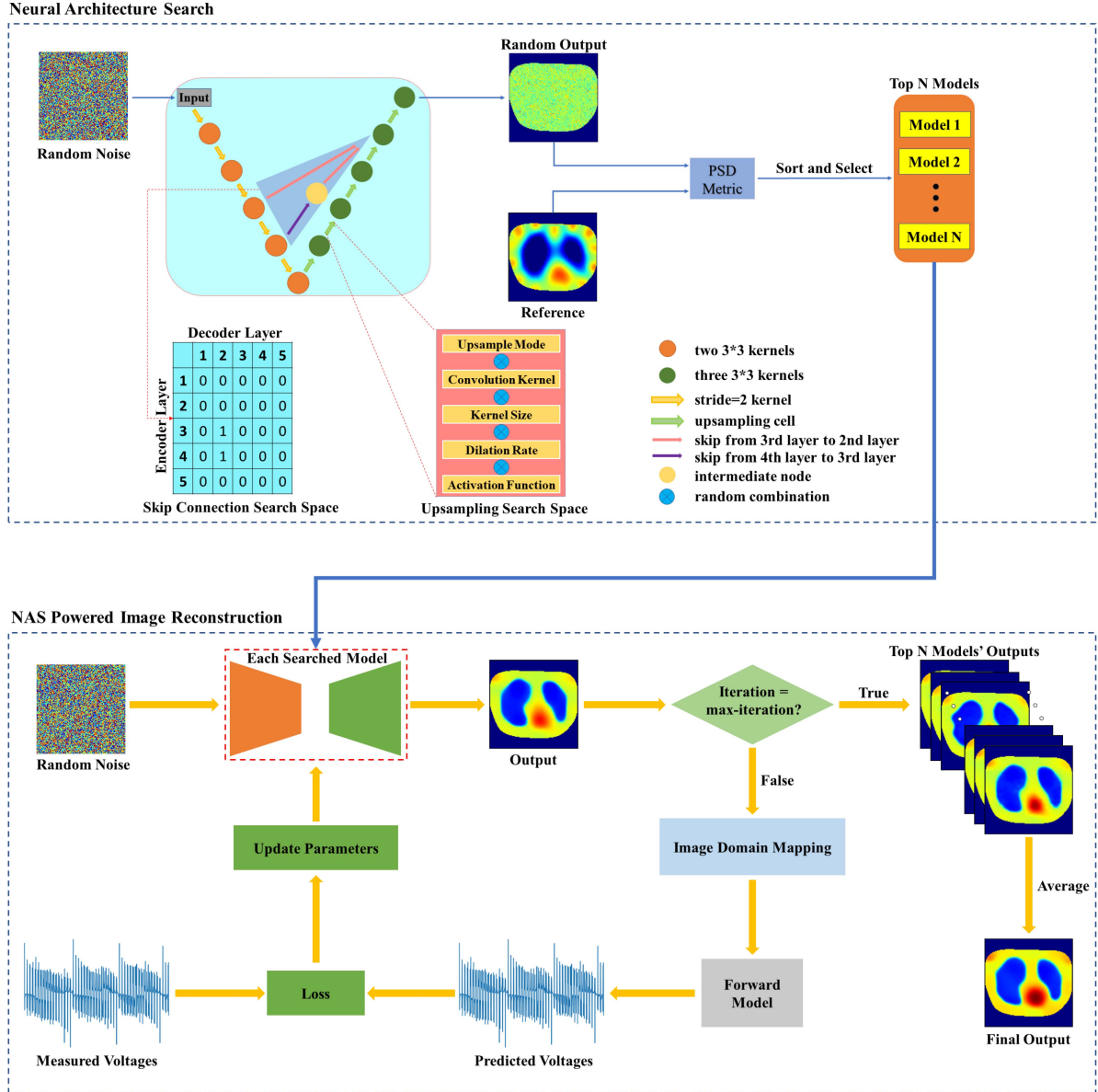


Fig. 1. The proposed methodology for reconstructing conductivity distribution using NAS techniques. The flowchart is divided into two main sections: the NAS Part (top) and the NAS powered Image Reconstruction Part (bottom). In the NAS Part, the search space includes upsampling and skip connection spaces, with shared weights denoted by rose-pink arrows. The selection of N models is determined by computing the PSD between the network's random output and the reference image. In the Reconstruction Part, the network takes random noise as input, undergoes continuous iterations with parameter updates guided by the loss function, and concludes when the stopping criteria are met, such as reaching the maximum predetermined iteration. The final output is the average result of the top N models' output.

Here, the variables  $v_x$  and  $v_y$  represent the frequency variables. Additionally,  $N_x$  and  $N_y$  denote the number of data points in the two frequency domains.

Following the definition provided in [29], the expressions for the power spectrum (PS) and PSD are as follows:

$$PS(v_x, v_y) = \frac{|F(v_x, v_y)|^2}{N_x^2 N_y^2}, \quad (13)$$

$$PSD(v_x, v_y) = \frac{PS(v_x, v_y)}{\Delta v_x \Delta v_y}, \quad (14)$$

where  $PS(v_x, v_y)$  is the power spectrum,  $PSD(v_x, v_y)$  is the power spectral density, and  $\Delta v_x$  and  $\Delta v_y$  represent the frequency increments.

This PSD distance serves as a benchmark, assigning scores to different models within the search space. Based on these scores, we can then select optimal architectures tailored to specific tasks. For a more detailed explanation of the calculations, please refer to Section IV-B.

#### D. EIT Reconstruction Framework With NASDIP

In this subsection, we address EIT inverse problems by combining NAS and DIP, incorporating an iterative-free approach known as the PSD metric for architecture selection. Our proposed method aims to harness the power of DIP while overcoming the complexities associated with manual network design. Unlike conventional NAS methods [30] [31], a notable

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**Algorithm 1:** NAS Powered DIP for EIT Reconstruction Method.

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1: Initialization and input: model NN input  $z$ , NN
   parameters  $\theta$ , learning rate  $lr$ , NOSER reconstruction
    $\sigma_{\text{NOSER}}$ , measured voltages  $V$ , domain mapping
   projection  $P$ , and number of max-iteration.
2: for each model NN in search space do
3:    $psd_z = \text{PSD}(\text{NN}(z; \theta))$ 
4:    $psd_{\text{NOSER}} = \text{PSD}(P(\sigma_{\text{NOSER}}))$ 
5:    $score = \text{MSE}(psd_z, psd_{\text{NOSER}})$ 
6: end for
7: Select top  $N$  models according to scores
8: for each selected model NN do
9:   repeat
10:     $\sigma = \text{NN}(z; \theta)$ 
11:     $U = U(P(\sigma))$ 
12:    Compute  $L = \|V - U\|^2 + \alpha TV(\sigma)$ 
13:    Compute  $\frac{\partial L}{\partial \sigma} = 2(\frac{\partial U}{\partial \sigma})^T (U - V) + \alpha \frac{\partial TV}{\partial \sigma}$ 
14:    Compute  $\frac{\partial L}{\partial \theta} = (\frac{\partial L}{\partial \sigma})^T \frac{\partial \sigma}{\partial \theta}$ 
15:    Update  $\theta$  using Adam optimizer
16:   until max-iteration
17: end for
18:  $\sigma_{\text{out}} = \frac{\sum_{i=1}^N \sigma_i}{N}$ 

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strength of our approach lies in its iterative-free nature during the architecture selection process.

The schematic representation of our proposed methodology is delineated in Fig. 1, with detailed elucidations provided in Section IV. After addressing the minimization problem posed in (9) or (10) via the chosen network determined by the PSD metric, the final reconstruction, namely the conductivity distribution, is attained. In the following, we provide the Pseudo code that encapsulates the execution steps of the proposed method.

#### IV. METHODS

In this section, we commence by offering a detailed overview of the search space. Following that, we delve into various types of PSD distance. In the next subsection, we furnish information on the test data and finite elements used in the validation studies. Lastly, we present implementation details regarding the evaluation of indicators, experimental settings, and the selection of hyperparameters.

##### A. Search Space

The search space in our methodology consists of two primary components: upsampling cells and skip connections. We utilize encoder-decoder neural networks, such as U-net, as a backbone due to their outstanding performance in addressing reconstruction issues.

As discussed in Section III-B, the upsampling cells' search space comprises five components: upsampling mode, convolution kernel for feature extraction, kernel size, dilation, and activation function. Spatial resolution can be changed using methods like bilinear upsampling, bicubic upsampling, nearest

neighbor interpolation, and the depth-to-space operation [32]. Notably, transposed convolution is avoided in upsampling cells to prevent checkboard artifacts [33]. After altering the feature size, options for feature processing include 2D convolution, depth-wise convolution [34], separable convolution, and identity. Different kernel sizes, ranging from  $1 \times 1$ ,  $3 \times 3$  to  $7 \times 7$ , are set for convolution kernels. Various activation functions, such as ReLU [35], LeakyReLU [36], SELU [37], and sometimes no activation function, are also incorporated. The entire upsampling search space is depicted in the rose-pink rectangular block in Fig. 1. The composition of diverse components ensures the ability of the proposed method to discover appropriate upsampling cells.

Unlike traditional skip connections, continuous links are split into separate connections. For instance, a skip connection transmitting feature information directly from the 1st layer to the 3rd layer is deconstructed into a connection from the 1st layer to the 2nd layer, followed by another connection from the 2nd layer to the 3rd layer. Additionally, each connection between the  $x$ -th layer and the  $(x+1)$ -th layer shares identical weights, represented by the two rose-pink straight arrows in Fig. 1. As shown in Fig. 1, the skip connection search space is a two-dimensional array of  $5 \times 5$ , where each number in the  $i$ -th row and  $j$ -th column indicates the presence or absence of skip connections between the  $i$ -th encoder layer and the  $j$ -th decoder layer (0 means no, and 1 means yes). This approach effectively reduces the dimensions of the search space and the number of parameters while maintaining the ability to transfer information.

With aforementioned search space and strategy, the proposed method can explore diverse neural architectures, enabling the selection of optimal architectures for different reconstruction problems. Moreover, the method achieves a balance between effectiveness and efficiency through weight sharing.

##### B. PSD Distance

To efficiently select the optimal architectures from the search space, we utilize PSD distance as a metric. The similarity in the power spectral densities (PSDs) of two signals suggests the presence of similar structural information in the signals. In other words, if the PSD of a model's output closely resembles the PSD of the reference conductivity distribution, it indicates the model's suitability for reconstructing that known conductivity distribution. Therefore, the distance between two PSDs serves as a benchmark for assigning scores to models within the search space.

The most straightforward approach involves calculating the mean square error of the two power spectral densities, using the logarithmic version known as decibel (DB) PSD to facilitate the calculation. This methodology, referred to as PSD DB MSE, is formulated as follows

$$\frac{1}{n} \sum_{x,y} (10 \cdot \log A_{x,y} - 10 \cdot \log B_{x,y})^2. \quad (15)$$

Here,  $A$  and  $B$  denote the PSD of the network's output and the conductivity distribution reconstructed by the NOSER method,

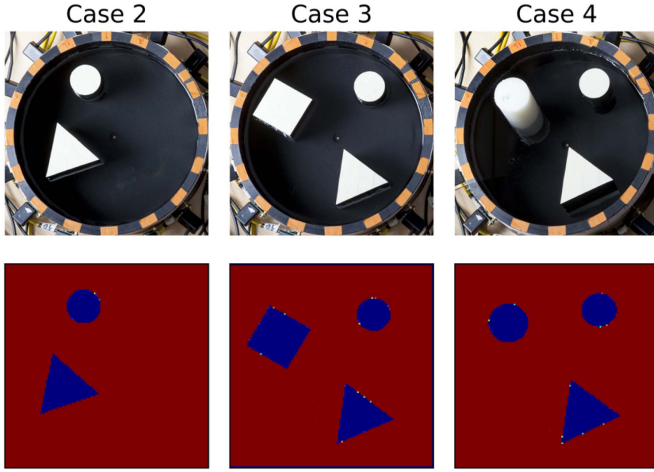


Fig. 2. Target photographs and masks utilized to generate the inclusion conductivity distribution for calculating performance metrics.

respectively. The parameter  $n$  represents the image size (e.g.,  $128 \times 128$ ).

In practice, when comparing images with and without noise, it becomes apparent that the low-frequency signal contains more information and less noise. Additionally, the low-frequency signal is concentrated around the center portion of the PSD image. Consequently, a ring-shaped mask is applied to the PSD to evaluate only the low-frequency subset of frequencies. This approach is commonly known as PSD DB Strip MSE:

$$\frac{1}{n} \sum_{x,y} (10 \cdot \log A_{x,y} - 10 \cdot \log B_{x,y})^2 \cdot M_{x,y}. \quad (16)$$

Here,  $A$  and  $B$  denote the PSD of the network's output and the conductivity distribution reconstructed by the NOSER method, respectively. The parameter  $n$  is the image size, and  $M$  is a binary mask consisting of 0s or 1s.

As an alternative metric, PSD Stripe Hist EMD introduces the conversion of PSD data into a histogram format, followed by the calculation of Earth Mover's Distance (EMD), as outlined in [28]. Our chosen methodology predominantly relies on the PSD DB Strip MSE approach for model selection. A comprehensive comparative analysis, employing the aforementioned metrics, is presented in Section VI-C.

### C. Test Data

We conducted validation for our method using both simulation data and experimental data. The simulation data was obtained from a simulated tank with a radius of 14 cm, featuring 16 equally spaced electrodes attached to the inner surface. The amplitudes of currents were set to 1 mA. In the numerical simulation, a forward mesh with 3154 nodes and 6130 elements was used to compute simulated measurements. For the inverse problem, the inverse mesh comprised 484 nodes and 902 elements. In Case 1 of Fig. 3, the conductivities of the heart, lungs, and background were set to 3 mS/cm, 0.5 mS/cm, and 2 mS/cm, respectively. In our experiments, Gaussian noise with a SNR of

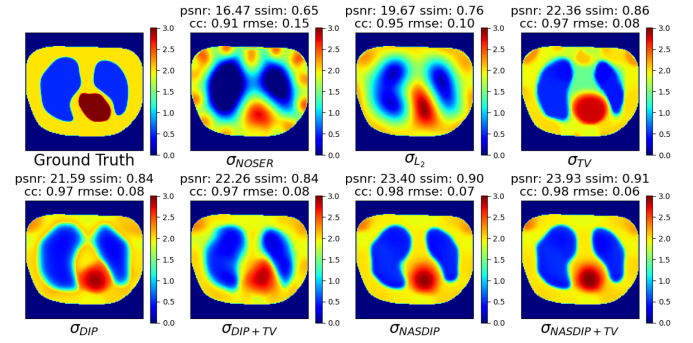


Fig. 3. Case 1: reconstructions with simulation human thorax data.

60 dB was added to the simulation data to simulate measurement errors.

The experimental data used during testing were sourced from the Open 2D Electrical Impedance Tomography dataset [38], acquired from the KIT4 measurement system [39]. The diameter of the cylindrical tank was 28 cm. Sixteen rectangular electrodes were attached equidistantly on the inner surface of the tank. The height of these electrodes was 7 cm and the width was 2.5 cm. Various inclusions were all plastic in experiments. The inverse mesh used for the experimental study was the same as the inverse mesh used in the simulation data.

### D. Implementation Details

In EIT reconstruction, as the ground truth of conductivity distribution is unavailable, we opt for the conductivity distribution reconstructed by the NOSER method as a reference. We employ a randomly initialized network, chosen from the search space, with random noise as input to generate an output without iteration. Subsequently, we calculate the PSD distance between this output and the reference distribution. Given the severely ill-posed nature of EIT, for improved robustness and stability to the proposed method, we opt for increased reliability by selecting the average of the top ten outputs (i.e.,  $N = 10$ ) as the final reconstruction. The size of the convolution kernel is set to  $3 \times 3$  in both encoders and decoders. As mentioned earlier, the adaptive moment estimation (Adam) method is utilized in our approach. The proposed method exhibits resilience to many of these hyperparameters, often allowing for adjustments within a broad range. Consequently, we choose the same learning rate  $lr = 0.001$  and weight  $\alpha = 0.001$  assigned to the TV regularization term as in DeepEIT [16] for fair comparison. Moreover, based on the differences between experimental and simulation cases, we set the maximum number of iterations to 2000 for experimental studies and 4000 for simulated test cases.

To quantitatively evaluate the reconstruction performance of the proposed method, we compute the following four metrics: the peak signal-to-noise ratio (PSNR) [40], the structural similarity index (SSIM) [41], the correlation coefficient (CC) [42], and the root mean square error (RMSE) [43]. In addition, to calculate the aforementioned evaluation criteria for the experimental studies with water tank data, real conductivity distributions are manually assigned based on the photographs, as depicted

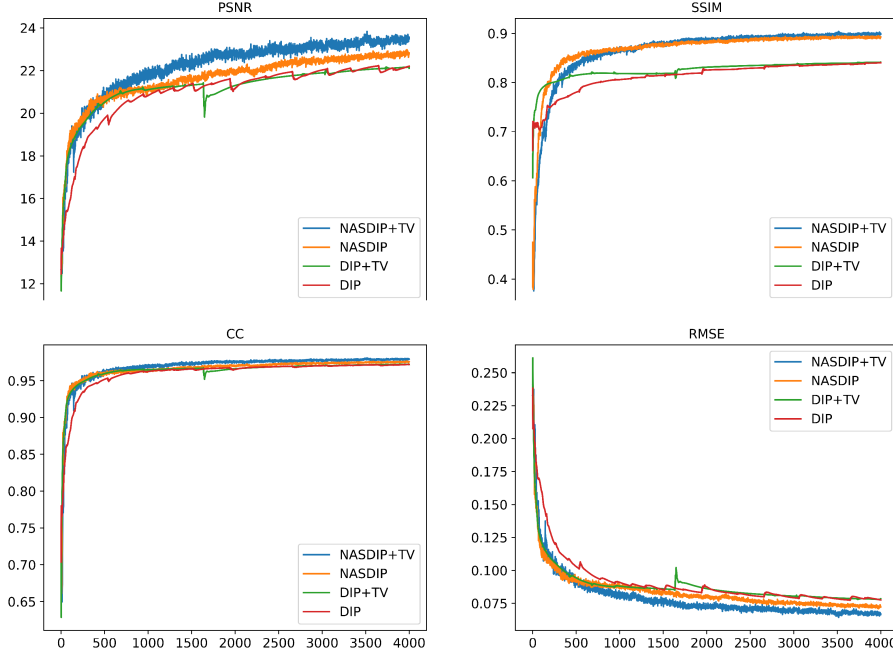


Fig. 4. Evaluation criteria of DIP-related reconstructions in Case 1 over iteration number.

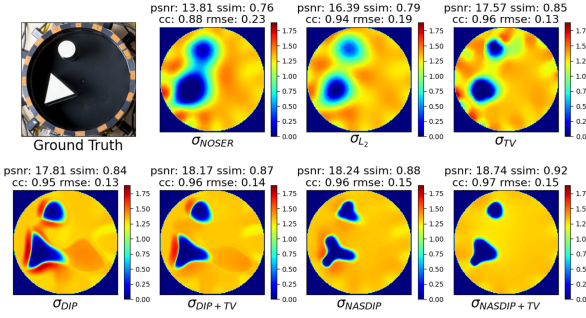


Fig. 5. Case 2: reconstruction results of experiment data with circular and triangular plastics.

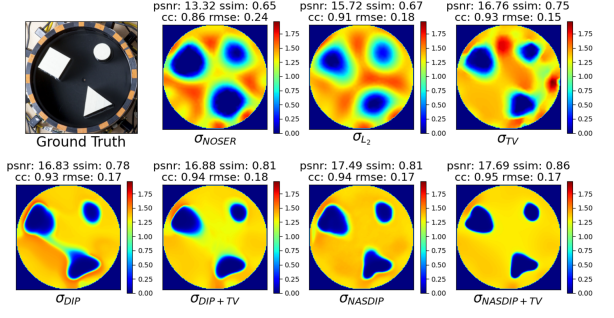


Fig. 6. Case 3: reconstruction results of experiment data with circular, triangular, and rectangular plastics.

in Fig. 2. It should be mentioned that our method does not require knowledge of the true conductivity distribution during the reconstruction process.

## V. RESULTS

To demonstrate the effectiveness of our method, we conduct a comparison with conventional reconstruction methods and the original DIP-based method. The conventional methods include the NOSER method [44], the  $L_2$  method based on smoothness prior, and the TV method [45] based on total variation prior. The first row of Fig. 3 and Figs. 5–7 in the comparison presents the ground truth alongside the reconstruction results obtained by NOSER,  $L_2$ , and TV methods. The second row of Fig. 3 and Figs. 5–7 showcases the reconstructions using four different convolutional neural network techniques: the original DIP-based method (marked as DIP), the original DIP-based method with TV prior (marked as DIP+TV), NASDIP, and NASDIP with TV prior (marked as NASDIP+TV). Alongside visual comparisons

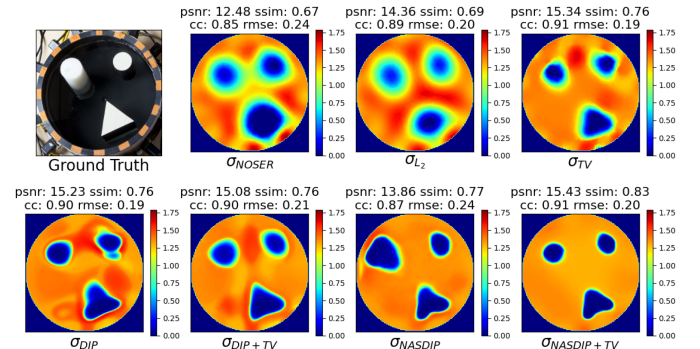


Fig. 7. Case 4: reconstruction results of experiment data with two circular and rectangular plastics.

of these methodologies, quantitative assessment criteria are also provided in these figures.

### A. Simulation Results

As depicted in Fig. 3, the NOSER approach struggles to effectively restore the heart, resulting in alterations to the contour of the lungs. Ring artifacts are observed surrounding the lungs, and the  $L_2$  approach incorrectly constructs the heart, exhibiting a noticeable artifact. The TV reconstruction result shows a staircase effect between the two lungs. Whereas the two original DIP-based approaches provide approximate reconstructions of tissue position and morphology, artifacts are slightly noticeable, and boundaries appear unclear. In contrast, the proposed method successfully restores the location and structure of the lungs, closely approximating the actual border. The high SSIM value of 0.91 indicates a significant similarity between the reconstructed structure and the real structure.

To assess the performance of DIP-related reconstructions, we plotted the evaluation criteria with respect to the iteration number in Fig. 4. Our observations indicate that NAS-enhanced DIP reconstructions exhibit superior performance, resulting in better evaluation values and less instability. It's worth emphasizing that the loss stabilization procedure (LSP) strategy [16], applied in the original DIP-based reconstructions, was eliminated in the current work.

### B. Experimental Results

Fig. 5 depicts the reconstructions of Case 2, where circular and triangular plastics were placed into the water tank. Whereas all methods can roughly locate the inclusions, the DIP-related reconstructions stand out with better distinguishability and enhanced structural information. It's worth noting that the original DIP-based methods while capturing shape information, tend to exaggerate the sizes of the inclusions and introduce artifacts with high conductivity around their boundaries. In contrast, the proposed NASDIP with and without TV outperforms conventional approaches and DIP-based methods, resulting in significantly fewer noticeable artifacts. This superiority is also reflected in the evaluation criteria presented in Fig. 5, where NASDIP+TV achieves the highest SSIM value of 0.92 and CC value of 0.97.

Next, let's delve into the reconstructions of Case 3 and Case 4, each featuring three plastic inclusions. In Figs. 6 and 7, the key distinction between Case 3 and Case 4 lies in the shape of the top-left inclusion within the saline tank. Upon a quick examination of these figures, it becomes evident that our proposed approach has produced the most favorable results. In fact, results obtained from three traditional methods offer only a rough indication of the inclusion locations, falling short of accurately determining their shapes and exhibiting distinct ring artifacts around them. Whereas the original DIP-based method outperforms conventional methods, noticeable artifacts still persist in the resulting images. Significantly, the proposed approach NASDIP, with and without TV, accurately portrays the location and shape of the inclusions. The evaluation criteria further support the superior performance of the proposed approach compared to alternative methods.

## VI. ABLATION AND ROBUSTNESS STUDIES

In this section, we perform ablation and robustness analyses to highlight the strengths of our approach. Firstly, we

elucidate the procedure for determining the optimal stopping point, considering the inherent instability in the learning process of conventional DIP. Subsequently, we explore the influence of TV regularization on our method. Following that, we assess the effectiveness of our approach using various PSD metrics. Finally, we showcase the robustness of our proposed method in the presence of noise.

### A. Optimal Stopping Point and Computational Efficiency

In conventional DIP-based procedures, the risk of over-fitting arises when reconstructing networks from a corrupted image. Early stopping techniques are commonly employed to address this issue, but their effectiveness depends on selecting the optimal stopping point, a challenging task in unsupervised network iteration. Our approach, as shown in Fig. 4, demonstrates sufficient stability, eliminating the need for explicit stopping point selection. Meanwhile, the iteration number were empirically set as 4000 for the simulation case and 2000 for the experimental cases. In this study, all the experiments are implemented in Windows PyTorch 1.10 using custom Python code on a desktop pc with an Intel i7-12700 CPU @ 2.10 GHz and 32 GB of RAM. Taking case 1 as an example, on average, the floating point operations per second of one searched model is about 10 Gflops and the computational time required for each searched model is approximately 50 minutes. It should be noted that the proposed method is formulated in the context of an absolute imaging framework, which is known to be time-consuming due to the repeated solving of the forward problem. A potential solution to speed up the reconstruction is to formulate the proposed method within the context of a difference imaging framework.

### B. TV Prior

The incorporation of TV prior in image reconstruction aims to preserve edge integrity without compromising smoothness. This significance is demonstrated in Fig. 4, where the blue curves representing our proposed method with TV regularization outperform the orange curves without TV regularization across four metrics. In Fig. 7, the comparison between our proposed method with and without TV regularization reveals that the boundaries of the three inclusions closely match the actual borders with TV regularization. Furthermore, the backgrounds of our proposed method, when employing TV regularization, exhibit a higher degree of uniformity. The evaluations in Figs. 3–7 highlight the substantial performance improvement resulting from the inclusion of TV regularization in both the DIP-based approach and our proposed method, showcasing a notable contrast with methods lacking TV regularization.

### C. Power Spectral Density Metrics

As discussed in Section IV-B, multiple metrics can be employed to calculate power spectral density distance, measuring the dissimilarity between two signals. To identify the most suitable statistic for our method, we compare reconstruction outcomes using multiple metrics in Case 1. Fig. 8 presents the ground truth of case 1, i.e., the conductivity distribution, on the left, followed by the outcomes of the reconstructions using

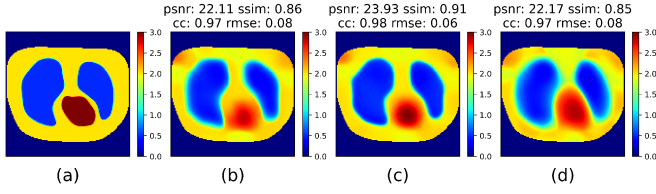


Fig. 8. Reconstruction outcomes utilizing various power density metrics. The first image on the left represents the ground truth (a), followed by the reconstruction results using PSD DB MSE (b), PSD DB Strip MSE (c), and PSD Strip Hist EMD (d) from left to right.

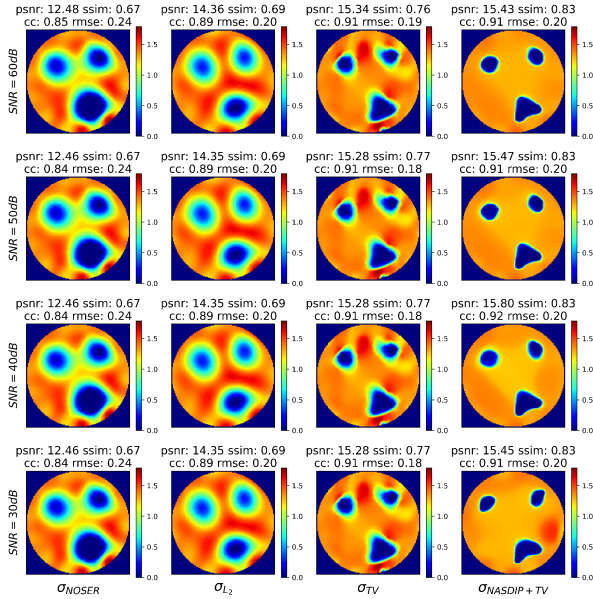


Fig. 9. Robustness study of the proposed approach against different noise levels.

three different metrics: PSD DB MSE, PSD DB Strip MSE and PSD Strip Hist EMD. Based on the visual comparisons and assessment indicators, we found that employing the PSD DB Strip MSE approach as the selection measure yields the most favorable results.

#### D. Robustness Study Against Different Noise Level

To comprehensively evaluate the robustness of our method in realistic scenarios, we introduced varying levels of noise to the experimental data in Case 4. Gaussian noise with SNRs ranging from 60 to 30 dB was added. Fig. 9 presents the reconstructions, with the first three columns showing outcomes from NOSER,  $L_2$ , and TV methods. The last column showcases the average outcomes using our proposed approach. Notably, our method consistently produces favorable results, even in the presence of Gaussian noise with an SNR as low as 30 dB, highlighting its resilience and competitiveness against noise.

### VII. CONCLUSION

This work introduces an innovative fusion of NAS with DIP to address the challenges inherent in Electrical Impedance Tomography (EIT) reconstruction. The method autonomously

navigates the search space to identify optimal neural architectures tailored to EIT reconstruction problems, thereby enhancing the adaptability and performance of DIP, which is inherently contingent on the input image. Notably, the proposed approach demonstrates robustness to noise, particularly in experimental settings, and alleviates the risk of overfitting during iterative processes. Leveraging random noise as an input, the selected neural networks contribute to the reconstruction of conductivity distributions. Through comprehensive simulation and experimental assessments, the method showcases superior performance compared to traditional techniques and the original DIP-based methods, affirming its efficacy in EIT reconstruction.

Future research avenues may explore the integration of additional learnable regularization terms beyond total variation to establish more comprehensive priors. Additionally, refining the model selection process by identifying the optimal model within the search space remains an area for further investigation, which could significantly improve the efficiency and utility of the proposed method in real-world applications.

In conclusion, our proposed method offers a promising solution for EIT reconstruction, outperforming current state-of-the-art unsupervised methodologies. The potential applicability of this approach extends to image reconstruction across diverse imaging modalities, underlining its significance in the broader context of medical imaging research.

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